

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 2nd Year 1st Term Examination-2021
Physical Chemistry
Ch 2119

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) What is vapour pressure? Show that the elevation of boiling point of a solution is directly proportional to the lowering of vapour pressure of that solution. 12
- 1(b) Describe Ostwald Walker method for the experimental determination of lowering vapour pressure. 12
- 1(c) A stream of dry air was passed through a solution containing 2.64g solute (an organic acid) in 30g solvent (diethyl ether) and then through pure solvent. The loss in weight of the solution was found to be 1.29g and that of ether was 0.069g. Calculate the molecular weight of organic acid. 11
- 2(a) Upon dilution, specific conductance decreases while equivalent conductance increases- explain. 08
- 2(b) Describe how Debye-Huckel theory. Explain the variation of equivalent conductance of strong electrolyte with concentration. 09
- 2(c) Why H^+ and OH^- show abnormal conductivity? Explain with suitable mechanism. 09
- 2(d) Whether $FeSO_4$ solution can be kept in copper pot or not? [Assume, $E^0_{Cu/Cu^{2+}} = 0.34V$ and $E^0_{Fe/Fe^{2+}} = -0.44V$]. 09
- 3(a) Why the presence of acid increase the rate but base decrease the rate of under-water corrosion-explain with the help of a diagram. 10
- 3(b) State Pilling Bedworth rule. Describe the function of anodic inhibitor in the prevention of corrosion on metal surface. 10
- 3(c) A sheet of carbon steel one meter wide by four meter long has lost 50g to corrosion over the past five months. Convert that mass loss to a penetration rate of the steel in mm units. 09
- 3(d) Is there any difference between corrosion and erosion? Explain. 06
- 4(a) Write down the electrode reactions during charging and discharging of Li-ion batteries with diagram. 13
- 4(b) What is meant by equivalent conductance at infinite dilution? How does Kohlrausch law help in determination of equivalent conductance of weak electrolyte at infinite dilution? 13
- 4(c) What are the basic characteristics of a solution? How do solutions differ from compounds? 09

SECTION-B

- 5(a) Distinguish between photochemical reaction and thermochemical reaction. 08
- 5(b) Reaction between H_2 and Cl_2 is a high quantum yield reaction on the other hand reaction between H_2 and Br_2 is a low quantum yield reaction. Why? Explain. 09
- 5(c) State and explain the law of photochemical equivalence and calculate the value of 1 einstein for light having $\lambda=2000\text{\AA}$. 10
- 5(d) Photodynamic therapy is an application of photochemistry in the treatment of cancer. Explain the mechanism to destroy the cancer cell. 08
- 6(a) On which process in between physisorption and chemisorption more energy is evolved and why? Explain. 08
- 6(b) Why does adsorption occur on solid or liquid surfaces? Explain with suitable diagram. 08
- 6(c) Write down the assumption of Langmuir adsorption isotherm and derive the equation pertaining to it and explain. 10
- 6(d) "Adsorption is an exothermic process" briefly illustrate this statement. 09
- 7(a) Compare dialysis and electrodialysis methods for the purification of colloidal solution. 08
- 7(b) How does positively and negatively charged AgI sol particles can be prepared? Explain. 09
- 7(c) Explain Hardy Schulze rule. Discuss the role of electrolyte on coagulation. 10
- 7(d) Describe some applications of colloids in our daily life. 08
- 8(a) Lyophobic sols can exhibit Tyndall effect whereas true solutions can not. Why? Explain. 09
- 8(b) Discuss the origin of charges on colloidal particles. 09
- 8(c) Define bio-luminescence and explain the mechanism of bio-luminescence occurred in the firefly system. 09
- 8(d) Derive an relation between quantum yield and rate of reaction. 08